

Performance Testing

Date	30 September 2026
Team ID	SWTID-2026-9357
Project Name	ComicCraftAI
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	JMeter
Type of Testing	Load Testing, Stress Testing, Spike Testing
Target Module / API	Login API / Comic Generation API
Test Environment	Cloud (AWS - EC2 Instance)
Test Date	30 September 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Registration (Concurrent Users)	50	300	Success rate $\geq$ 95% Response time < 2 sec
2	Login API (Concurrent Users)	100	300	Success rate $\geq$ 95% Response time < 2 sec
3	Comic Generation API (Concurrent Users)	50	600	Success rate $\geq$ 90% Response time < 5 sec
4	Load Test (Peak Load) (Overall System)	200	900	System stable No major errors Response time < 5 sec

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.8 seconds	Pass	Within target
2	Response Time (Max)	< 5 seconds	4.6 seconds	Pass	Within target
3	Throughput (Req/sec)	-	8.5 req/sec	Pass	Stable throughput
4	Error Rate	< 1%	0.5%	Pass	Low error rate
5	CPU Utilization	< 80%	68%	Pass	Within limit
6	Memory Utilization	< 80%	72%	Pass	Within limit

Step 4: Observations & Analysis

Key Findings:

System remained stable under expected and peak load; response times stayed within target limits.

Bottlenecks Identified:

Comic generation API showed higher response time under peak load.

Optimization Steps Taken:

Reduced unnecessary processing, optimized API calls, and improved resource handling.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

